

Daniel S.W. Lee, Ph.D.

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EDUCATION

PRINCETON UNIVERSITY <i>PhD</i> , Quantitative and Computational Biology	Princeton, NJ 2021
PRINCETON UNIVERSITY <i>MA</i> , Quantitative and Computational Biology	Princeton, NJ 2018
YALE UNIVERSITY <i>BS</i> , Physics; Molecular Biophysics and Biochemistry	New Haven, CT 2016

RESEARCH EXPERIENCE

CALIFORNIA INSTITUTE FOR QUANTITATIVE BIOSCIENCES, U.C. BERKELEY <i>Postdoctoral Scholar</i> Advisor: Prof. Dan Fletcher	Berkeley, CA Apr 2022 - Present
LEWIS-SIGLER INSTITUTE FOR INTEGRATIVE GENOMICS, PRINCETON UNIVERSITY <i>Postdoctoral Research Associate</i> Advisor: Prof. Ned Wingreen	Princeton, NJ Nov 2021 - Apr 2022
LEWIS-SIGLER INSTITUTE FOR INTEGRATIVE GENOMICS, PRINCETON UNIVERSITY <i>PhD Student</i> Advisors: Profs. Cliff Brangwynne and Ned Wingreen	Princeton, NJ Sept 2016 - Nov. 2021
DEPT. OF BIostatISTICS, YALE SCHOOL OF PUBLIC HEALTH <i>Research Assistant</i> Advisor: Prof. Jeffrey Townsend	New Haven, CT Jan. 2016 - Aug. 2016
DEPT. OF MOLECULAR BIOPHYSICS AND BIOCHEMISTRY, YALE UNIVERSITY <i>Undergraduate Research Student</i> Advisor: Prof. Jonathon Howard	New Haven, CT Aug 2014 - Aug 2016
SCHOOL OF LIFE SCIENCES, TSINGHUA UNIVERSITY <i>Research Assistant</i> Advisor: Prof. Xin Liang	Beijing, China May 2015-Aug 2015
MAX PLANCK INSTITUTE FOR CELL BIOLOGY & GENETICS <i>Undergraduate Research Student</i> Advisor: Prof. Jonathon Howard	Dresden, Germany May 2014-Aug 2014
DEPT. OF MOLECULAR BIOPHYSICS AND BIOCHEMISTRY, YALE UNIVERSITY <i>Research Assistant</i> Advisor: Prof. Patrick Sung	New Haven, CT May 2013 - May 2014

AWARDS & HONORS

BURROUGHS WELLCOME FUND <i>Career Awards at the Scientific Interface</i>	JULY 2026-JULY 2031
NATIONAL SCIENCE FOUNDATION GRADUATE RESEARCH FELLOW <i>NSF-GRFP in Physics and Astronomy, Physics of Living Systems</i>	SEPT. 2018- MAY 2021

PUBLICATIONS

2494 citations, $h=13$ (from Google Scholar, current as of 08/18/2026)

*indicates equal contribution

† indicates corresponding author

Lee, D.S.W.*, Oster, L.F*, Son, S., Fletcher, D.A.† **Cell surface crowding is a tunable energetic barrier to cell-cell fusion.** *Nature Communications* 16 (1), 7158 (2025).

Choi, C.-H., Lee, D.S.W., Sanders, D.W., Brangwynne, C.P.† **Condensate interfaces can accelerate protein aggregation.** *Biophysical Journal* 123(11), 1404-1413 (2024).

Takatori, S.C.*†, Son, S.*†, Lee, D.S.W., Fletcher, D.A. † **Engineered molecular sensors of cell surface crowding.** *Proceedings of the National Academy of Sciences* 120 (21), e2219778120 (2023).

Lee, D.S.W., Choi, C.-H., Riback, J.A., Sanders, D.W., Beckers, L., Brangwynne, C.P.†, Wingreen, N.S.† **Size distributions of intracellular condensates reflect competition between coalescence and nucleation.** *Nat. Phys.* **19**, 586–596 (2023)

Riback, J.A.*, Eeftens, J.M.*, Lee, D.S.W.*, Beckers, L.A., Becker, L., Sanders, D.W., Wingreen, N.S., Brangwynne, C.P.† **Viscoelastic RNA entanglement and advective flow underlie nucleolar form and function.** *Molecular Cell* **83**, 3095-3107 (2023).

Lee, D.S.W., Strom, A.R., Brangwynne, C.P.†, **The mechanobiology of nuclear phase separation.** *APL Bioengineering*. 6(2), 021503 (review, 2022)

Jack A.*, Kim, Y.*, Strom, A.R.*, **Lee, D.S.W.**, Williams, B., Schaub, J.M., Kellogg, E.H., Finkelstein, I.J., Ferro, L.S.†, Yildiz, A.†, Brangwynne, C.P.†, **Compartmentalization of Telomeres through DNA-scaffolded Phase Separation.** *Developmental Cell* 57 (2), 277-290. e9 (2022).

Zhang, Y.*, **Lee, D.S.W.***, Meir, Y., Brangwynne, C.P., Wingreen, N.S.†, **Mechanical frustration of phase separation in the cell nucleus by chromatin.** *Physical Review Letters.* **126**, 258102 (2021).

Lin, C.K.*, **Lee, D.S.W.***, McKeithen-Mead, S., Emonet, T., Kazmierczak, B.†, **A primed subpopulation of bacteria allows rapid expression of the T3SS in *Pseudomonas aeruginosa*.** *mBio* 12(3), e00831-21 (2021).

Lee, D.S.W., Wingreen, N.S.†, Brangwynne, C.P.† **Chromatin mechanics dictates subdiffusion and coarsening dynamics of embedded condensates.** *Nat. Phys.* **17**, 531-538(2021). Featured on April 2021 cover.

Sanders, D.W., Kedersha, N.*, **Lee, DSW.***, Strom, A.R.*, Drake, V., Riback, J.A., Bracha, D., Eeftens, J.M., Iwanicki, A., Wang, A., Wei, M.T., Whitney, G., Lyons, S.M., Anderson, P., Jacobs, W.M., Ivanov, P., Brangwynne, C.P.† **Competing protein-RNA interaction networks control multiphase intracellular organization.** *Cell* 181 (2), 306-324. e28 (2020).

Shin Y.*, Chang Y.-C.*, **Lee, DSW.**, Berry, J., Sanders, D.W., Ronceray, P., Wingreen, N.S., Haataja, M., Brangwynne, C.P.† **Liquid nuclear condensates mechanically sense and restructure the genome.** *Cell* 175 (6), 1481-1491, e13 (2018).

Zhao, Z.M.*, Campbell, M.C.*, Li, N., **Lee, D.S.W.**, Zhang, Z., Townsend, J.P.† **Detection of regional variation in selection intensity within protein-coding genes using DNA sequence polymorphism and divergence.** *Molecular Biology and Evolution*, 34 (11), 3006-3022 (2017).

PROFESSIONAL ACTIVITIES & SERVICE

JOURNAL REFEREE

2024-PRESENT

Reviewed manuscripts for *PRL*, *PRE*, and *PRX Life*

CLAUDE SCIENCE COHORT

SEPT-DEC 2026

Selected as 1 of 50 teams for the inaugural Claude Science Cohort to receive \$30,000 in API credits and \$2000 in Modal compute for project "Discovering strategies to manipulate cell surface accessibility" using AI to link gene expression to cell surface biophysics.

MENTORSHIP

Mentored undergraduate Alan Amin (summer 2018) in project investigating mechanical perturbation of chromatin in living cells, undergraduate Angela Yang (summer 2020) in developing and analyzing polymer simulations in Lammmps molecular dynamics package, and Molecular Biology rotation student Jordy Botello in high-throughput image analysis project (2021), undergraduate Natalia Orlovsky in project on subdiffusive fluctuations in the nucleus (2021).

Mentored Fletcher lab rotation students Maddie Blaauw and Kira Buttrey (UCB BioE program) in developing oligo-based cell surface crowding probes; undergraduate Yuri Onimura (Molecular Cell Biology major), in project analyzing CRISPR screening readouts; intern Yue See Lee (California Institute for Regenerative Medicine program through Berkeley City College), in project on myoblast fusion.

MCGRAW TEACHING TRANSCRIPT

Participated in McGraw teaching transcript program, focusing on training pedagogical skills. Activities include attending workshops, designing teaching statement and sample syllabus and receiving feedback on class observations.

WOODS HOLE PHYSIOLOGY COURSE 2019, 2023, 2024

SUMMER 2019, 2023, 2024

Teaching Assistant. Designed and administered tutorials and independent projects for students in Woods Hole Physiology Course 2019 as part of Brangwynne lab and in 2023 and 2024 as part of Fletcher lab.

ASSISTANT IN INSTRUCTION

FALL 2018, 2019

QCB 515, Methods and Logic in Quantitative and Computational Biology. Designed problems and lessons to instruct interdisciplinary group of students in computational and theoretical approaches to biology under Prof. Ned Wingreen.

2017 CARGESE SUMMER SCHOOL ON THEORETICAL BIOPHYSICS

JUNE - JULY 2017

Student

Participated in two-week summer school designed to instruct graduate students in theoretical methods and approaches to biophysics held in Corsica, France, culminating in original research projects. Supervised by Thierry Mora, Aleksandra Walczak, and Ilya Nemenman.

YALE UNDERGRADUATE RESEARCH ASSOCIATION

JAN. 2014-MAY 2016

Founder & Director

Founded organization to build and serve community among undergraduates pursuing research at Yale.

CONFERENCES & TALKS

Lee, D.S.W., Oster, L.F., Son, S.S., Fletcher, D.A.. **Cell Surface Crowding is a Biophysical Barrier to Cell-cell Fusion.** American Society of Cell Biology 2024 meeting, Subgroup talk

Lee, D.S.W., Kim, J.Y., Robey, E.A., Fletcher, D.A. **Regulation of cell surface crowding on cancer cells and immune cells.** American Society of Cell Biology 2023 meeting, Subgroup talk

Lee, D.S.W., Brangwynne, C.P., Wingreen, N.S., **Physical Principles Underlying Intracellular Condensate Size Distributions.** Invited seminar, **NYU Chromatin Club.** September 29, 2023.

Lee, D.S.W., Brangwynne, C.P., Wingreen, N.S., **Physical Principles Underlying Intracellular Condensate Size Distributions.** APS March Meeting 2022.

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Chromatin Mechanics Dictates Subdiffusion and Coarsening Dynamics of Embedded Condensates.** American Society of Cell Biology 2020 Virtual Meeting, Subgroup talk

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Subdiffusive Dynamics of Optogenetic Droplets Measure Chromatin Mechanics.** Invited talk, **NSF-Funded MIT Genome Architecture and Function Virtual Workshop 2020.**

Lee, D.S.W., Chang, Y.-C., Shin, Y., Sanders D.W., Bracha, D., Wingreen, N.S., Brangwynne, C.P. **Mechanical Interplay of Chromatin and Liquid-Liquid Phase Separated Condensates.** APS March Meeting 2019, Boston, M.A.

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Subdiffusive Dynamics of Optogenetic Droplets Measure Chromatin Mechanics.** 2018 4DN-ASCB Satellite Meeting, San Diego, C.A., Lightning Talk & Poster

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Dynamic Organization of Chromatin by Liquid-Liquid Phase Separation.** 2018 National Human Genome Research Institute Training and Career Development Meeting, Los Angeles, C.A., Lightning Talk